

HOTEL MANAGEMENT SYSTEM A C PROGRAMMING PROJECT

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ABSTRACT

This project presents a basic console-based Hotel Management System developed using the C programming language. It allows hotel staff to efficiently manage guest information, assign rooms, track meal preferences, handle buffet services, and record payment methods. The system utilizes modular programming concepts including functions, structures, and header/source file separation. It provides a simple interactive menu and supports operations such as adding guest details, displaying records, assigning rooms, and calculating basic charges. Although limited to basic functionality, the application demonstrates core programming logic and structured project design suitable for learning and academic purposes.

1. PROBLEM DEFINITION

Manual hotel data handling is inefficient and prone to errors. This project aims to automate basic hotel operations by designing a console-based system that:

- Records and stores guest details
- Assigns rooms
- Tracks food preferences and services
- Calculates costs and basic billing
- Presents a menu-driven user interface

2. System Design

Flowchart: START → Display Menu → Option Selected → (Add Guest / Display Guests / Assign Room / Services / Exit) → Execute Function → Return to Menu → END

Algorithm Example (Add Guest):

1. Start
2. Prompt user to enter guest ID, name, room number, meal type, service type, buffet type, and payment method
3. Store values in relevant structure
4. Display confirmation
5. Stop

3. IMPLEMENTATION

DETAILS

- Programming Language: C •
- Compiler: GCC (MinGW) • IDE: Visual Studio Code • Project Structure: hotel-management-project/

- src/ (main.c, guest.c, room.c, services.c)
 - include/ (guest.h, room.h, services.h)
 - docs/ (ProjectReport.pdf) README.md

4.TESTING & OUTPUT

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

hotel + - [] [x] ... [refresh] [x]

5. Exit
Enter your choice: 1

===== ADD GUEST DETAILS =====
Enter Guest ID: 23455
Enter Guest Name: rishabh
Enter Room Number: 19
Meal Type (veg/nonveg): nonveg
Service Type (room/self): room
Buffet Type (breakfast/lunch/dinner): breakfast
Payment Method (cash/card/upi): upi

New Guest added successfully!

===== HOTEL MANAGEMENT SYSTEM =====
1. Add Guest Details
2. Display All Guests
3. Assign Room
4. Food & Services
5. Exit
Enter your choice: 2

===== HOTEL GUEST LIST =====

ID	Name	Room	Meal	Service	Buffet	Payment
23455	rishabh	19	nonveg	room	breakfast	upi

===== HOTEL MANAGEMENT SYSTEM =====
1. Add Guest Details
2. Display All Guests
3. Assign Room
4. Food & Services
5. Exit
Enter your choice: █

Build with Agent

AI responses may be inaccurate.
[Generate Agent Instructions](#) to onboard AI onto your codebase.

🔗 C guest.c +

Describe what to build next

Agent ▾ Auto ▾ 🧑🏿🧑🏻🧑🏾🧑🏼

🔄 ➡

src > C guest.c > [icon] displayGuests()

13 void addGuest() {
14 printf("Meal Type (veg/nonveg): ");
15 scanf("%s", guests[totalGuests].meal_type);
31
32 printf("Service Type (room/self): ");
33 scanf("%s", guests[totalGuests].service_type);
34
35 printf("Buffet Type (breakfast/lunch/dinner): ");
36 scanf("%s", guests[totalGuests].buffet_type);
37
38 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

hotel + - [] [x] ... [refresh] [x]

Room No : 19
Room Type : deluxe
AC Type : ac
Days of Stay : 1
Total Room Charge: 10000.00

===== HOTEL MANAGEMENT SYSTEM =====
1. Add Guest Details
2. Display All Guests
3. Assign Room
4. Food & Services
5. Exit
Enter your choice: 4

===== HOTEL SERVICES (FOOD) =====
Enter Room Number: 19
Meal Type (veg/nonveg): nonveg
Buffet Type (breakfast/lunch/dinner): lunch
Room Service? (yes/no): yes

----- SERVICE SUMMARY -----
Room Number : 19
Meal Type : nonveg
Buffet Type : lunch
Room Service : yes
Total Charge : 600.00

===== HOTEL MANAGEMENT SYSTEM =====
1. Add Guest Details
2. Display All Guests
3. Assign Room
4. Food & Services
5. Exit
Enter your choice: █

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Agent ▾ Auto ▾ 🧑🏿🧑🏻🧑🏾🧑🏼

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5. CONCLUSION

The Hotel Management System successfully meets basic operational requirements. It showcases academic-level implementation of structured programming in C and demonstrates concepts such as modularity, functions, and file organization. The system is functional and may be extended with additional features like data persistence and GUI in future implementations.

6. FUTURE SCOPE

- Add file handling for data storage
- Implement login and authentication
- Create GUI version
- Integrate billing system with tax calculation
- Support multiple guest check-ins in one session

7. REFERENCES

• Let Us C – Yashwant Kanetkar • C Programming Documentation •
TutorialsPoint.com • Visual Studio
Code Documentation • GCC Official
Documentation